

**Pokhara University**  
**Gandaki College of Engineering and Science**

Lamachaur, Kaski



**Lab 1: Implementation of Source Code Management Using Git  
Commands Through Github**

***Submitted By:***

Sajish Bastola

Roll no. 32

***Submitted To:***

Er. Rajendra Bahadur Thapa

## Lab Report

### Objective

To understand and apply Source Code Management (SCM) using Git commands and GitHub for version control and project management.

### Tools Required

- Git
- GitHub Account
- Git Bash / Terminal
- Visual Studio Code (Optional)

### Theory

Source Code Management (SCM) is the process of tracking and managing changes to source code. Git is a distributed version control system that records changes, while GitHub is a cloud-based platform for hosting and collaborating on Git repositories.

### Procedure

1. Install Git and configure the username and email.
2. Create a project folder and initialize a Git repository.
3. Add project files to the staging area.
4. Commit the changes to the local repository.
5. Create a new repository on GitHub.
6. Link the local repository to the GitHub repository.
7. Push the local repository to GitHub.

### Git Commands (Coding Section)

#### # Configure Git

```
git config --global user.name "Your Name"  
git config --global user.email "your@email.com"
```

#### # Create and initialize repository

```
mkdir MyProject  
cd MyProject  
git init
```

#### # Create a sample file

```
echo "Hello GitHub!" > README.md
```

#### # Stage and commit changes

```
git add .  
git commit -m "Initial commit"
```

#### # Connect to GitHub repository

```
git remote add origin https://github.com/username/MyProject.git
```

```
# Push project to GitHub  
git branch -M main  
git push -u origin main
```

```
# Check repository status  
git status
```

```
# View commit history  
git log
```

### **Output**

- A local Git repository was successfully created.
- Files were tracked and committed.
- The repository was connected to GitHub.
- The project was successfully pushed to the remote GitHub repository.

### **Result**

Successfully implemented Source Code Management using Git commands and synchronized the local repository with GitHub.

### **Conclusion**

This lab demonstrated the complete Git workflow, including repository initialization, staging, committing, connecting to GitHub, and pushing code. It provided practical experience with version control and collaborative software development.